

WESTMINSTER INTERNATIONAL UNIVERSITY IN TASHKENT

COMPUTER LAB TEST - Slot 1

SEMESTER 1

November 2025

MODULE CODE: 5BUIS009C
MODULE TITLE: Database Systems
DATE: 2025-11-10
TIME: 9:00
VENUE: SHB408

INSTRUCTIONS TO CANDIDATES:

ANSWER ALL QUESTIONS

You should open MSSQL Server Management Studio and work with Northwind database.

Save your answers in ONE file with name **YourID_DBSTest.txt**

Upload file to the Intranet.

TIME ALLOWED: ONE HOUR and FIFTEEN MINUTES

**PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED
TO DO SO BY THE INVIGILATOR**

Slot 1

Using Northwind MS SQL Server database, write SQL queries to solve the following tasks.

Question 1 (8 marks)

Write a query to retrieve all product names and unit prices from the Products table where the unit price is greater than \$20, ordered by product name in ascending order.

Use tables: Products.

Question 2 (12 marks)

Write a query to find all customers from Germany, UK, or France whose company name starts with 'A' or 'B' and has more than 15 characters in the company name. Display company name, contact name, and country, ordered by the length of the company name in descending order.

Use tables: Customers.

Question 3 (14 marks)

Write a query to show order details including customer company name, employee's full name (first and last), and order date for all orders placed in the first quarter of 1997. Display results ordered by employee last name and then by order date.

Use tables: Orders, Customers, Employees.

Question 4 (14 marks)

Write a query to find categories that have more than 8 products, showing the category name, total number of products, and average unit price. Only include categories where the average unit price is between \$15 and \$50. Order results by the number of products in descending order.

Use tables: Products and Categories.

Question 5 (16 marks)

Write a query to find all products that belong to the same category as 'Chai' (but exclude 'Chai' itself), showing the product name and the total number of times each product has been ordered using a correlated subquery.

Use tables: Products and Order Details.

Slot 1

Question 6 (22 marks)

Write a query showing order details with customer company name, employee full name, product name, supplier company name, and order date. Include window functions to show: total number of orders per customer, earliest order date per customer, and total count of all orders in the system. Order by customer company name and order date.

Use tables: Products, Suppliers, Orders, Order Details, Customers, Employees.

Question 7 (14 marks)

Optimise one of the queries that you have produced for question 2-6 by suggesting relevant indexes that will improve its performance. Create suggested indexes and check if they are actually used and thus improve the performance. Critically discuss your choice of indexes, support your answer with reference to query execution plans. Your answer should include:

- The SQL script for creating the indexes.
 - The execution plans before and after index creation in plain text format (use `set showplan_text ON / OFF`).
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